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Linearization of airflow through zone dampers of an HVAC system

Abstract

Embodiments include systems and methods for linearization of airflow through dampers of a heating, ventilation, and air conditioning systems. In one embodiment, a system may include a fan, a first damper for a first zone of the system, the first damper configured to move from a first position to a second position, and from the second position to a third position; and a controller. The controller may be configured to determine a first corrective positional adjustment for the first damper for the second position at a first time interval, and determine, using the first corrective positional adjustment, a fourth position for the first damper, wherein the first damper moves to the fourth position instead of the second position during a second time interval.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/750,259, filed May 20, 2022 (issued as U.S. Pat. No. 11,703,249), which is a continuation of U.S. application Ser. No. 16/866,032, filed May 4, 2020 (issued as U.S. Pat. No. 11,339,991), which is a continuation of U.S. application Ser. No. 15/850,609, filed Dec. 21, 2017 (issued as U.S. Pat. No. 10,641,515), each of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(1) The present disclosure relates generally to temperature control systems, and more particularly to linearization of airflow through zone dampers of a heating, ventilating, and air-conditioning (HVAC) system.

BACKGROUND

(2) Temperature control systems, such as multi-zone HVAC systems (hereinafter ‘HVAC system’) include one or more components to condition air that enters the system and drive the conditioned air through supply ducts to multiple zones within a building. Each supply duct includes zone dampers that may be adjusted to control a flow of the conditioned air into each zone to achieve a desired temperature within the zone. Some zone dampers, such as modulating zone dampers, may include intermediate positions between a fully open position and a fully closed position such that the zone dampers can be incrementally closed to achieve a desired flow of the conditioned air in a zone. The number of available intermediate positions may vary based on a desired granularity in controlling the flow of the conditioned air to the zone.

(3) Conventional HVAC systems operate under the assumption that the relationship between a change in the damper positions of the zone dampers of the HVAC system and the volume of conditioned air flowing through the zone dampers is a linear relationship. The linear relationship between the flow of the conditioned air (hereinafter ‘airflow’) with each change in damper position of the zone damper is desired for obtaining the best performance in the HVAC system. However, typically, most zone dampers exhibit nonlinear behavior of airflow with the change in damper positions of the zone damper, which in turn causes rough and uneven changes in temperature and airflow, excess airflow noise, and/or inadequate conditioning of the zone. The nonlinear behavior of the airflow may also cause the HVAC system to suffer from unexpected perturbations and even go out of control.

(4) In light of the above mentioned shortcomings of conventional HVAC systems, zone dampers that exhibit a linear behavior of airflow through the zone dampers for each damper position are described herein. It is noted that this background information is provided to reveal information believed by the applicant to be of possible relevance to the present disclosure. No admission is necessarily intended, nor should be construed, that any of the preceding information constitutes prior art against the present disclosure.

SUMMARY

(5) In one aspect, the present disclosure relates to a control system to obtain a linear behavior of airflow through a zone damper of a zone of an HVAC system with a change in damper positions of the zone damper. The control system includes an air handler that is configured to deliver the airflow through the HVAC system. The control system further includes the zone damper that is configured to adjust the airflow to the zone of the HVAC system, wherein the damper positions of the zone damper comprises at least one intermediate position between a fully open position and a fully closed position. Furthermore, the control system includes a system controller that is coupled to the air handler and the zone damper. The system controller is configured to instruct the air handler to maintain a fixed airflow through the HVAC system. Further, while maintaining a fixed airflow through the HVAC system, the system controller is configured to record a static pressure across the HVAC system when the zone damper is in: (a) the fully open position, (b) the at least one intermediate position, and (c) the fully closed position. Furthermore, the system controller is

configured to determine a corrected intermediate position by applying a correction to the at least one intermediate position for obtaining the linear behavior of airflow through the zone damper. The corrected intermediate position is determined based on the static pressure when the zone damper is in the fully open position, the static pressure when the zone damper is in the at least one intermediate position, and the static pressure when the zone damper is in the fully closed position. The system controller is configured to store the corrected intermediate position in a memory of the system controller. The corrected intermediate position that is stored in the memory of the system controller is used to adjust a position of the zone damper during an operational phase of an HVAC system.

(6) In another aspect, the present disclosure relates to a system controller of an HVAC system. The system controller includes a processor, and a memory that comprises instructions for obtaining a linear behavior of airflow through a zone damper of a zone of the HVAC system with a change in damper positions of the zone damper. When the instructions are executed by the processor, the instructions cause the processor to control an air handler of the HVAC system to maintain a fixed airflow through the HVAC system, and turn off one or more temperature control elements of the HVAC system. Further, for each zone damper, the instructions cause the processor to control incrementally close the zone damper by sequentially progressing the zone damper through a plurality of intermediate positions between a fully open position and a fully closed position. Furthermore, for each intermediate position of the zone damper, the instructions cause the processor to record a static pressure across the HVAC system when the zone damper is in: (a) the fully open position, (b) the intermediate position, and (c) the fully closed position; determine a corrected position associated with the intermediate position for obtaining the linear behavior of airflow through the zone damper, and store the corrected position in the memory of the system controller, wherein the system controller uses the corrected position that is stored in the memory to adjust a damper position of the zone damper during an operational phase of an HVAC system. The corrected position is calculated using a mathematical model comprising the following mathematical equations:

$$\text{Corrected damper position}_n = (\text{Damper position}_n - (1 - \text{Correction percent}_n) * (\text{Damper position}_n - \text{Damper position}_{(n+1)})),$$

$$\text{Correction percent}_n = ((Kn_{\text{ideal}} - K_{(n+1)})) / ((Kn - K_{(n+1)})), \text{ and}$$

$$Kn = ((1 - \sqrt{SP_{\text{position}}(n) / (SP_{\text{zone closed}})}) / (1 - \sqrt{SP_{\text{open}} / (SP_{\text{zone closed}})})),$$

where SP_{position} is a value of the static pressure across the HVAC system when the zone damper is the intermediate position, $SP_{\text{zone closed}}$ is a value of the static pressure across the HVAC system when the zone damper is in the fully closed position, SP_{open} is a value of the static pressure across the HVAC system when the zone damper is in the fully open position, Kn_{ideal} is the value of a system constant associated with the intermediate position in an ideal system with the linear behavior, Kn is a value of the system constant that is calculated based on the values of the static pressure when the zone damper is in the fully open position, the intermediate position, and the fully closed position, $K_{(n+1)}$ is a value of a system constant associated with a damper position of the zone damper that sequentially follows the intermediate position, Damper position_n is the intermediate position of the zone damper, and $\text{Damper position}_{(n+1)}$ is the damper position of the zone damper that sequentially follows the intermediate position.

(7) In yet another aspect, the present disclosure relates to a method of a control system for obtaining linear behavior of airflow through a zone damper of a zone of an HVAC system with a change in damper positions of the zone damper. The method includes instructing an air handler of the HVAC system to maintain a fixed airflow through the HVAC system. Further, while maintaining a fixed airflow through the HVAC system, the method includes recording a static pressure across the HVAC system when the zone damper is in: (a) a fully open position, (b) an intermediate position between the fully open position and a fully closed position, and (c) the fully closed position. Furthermore, the method includes determining a corrected intermediate position by

applying a correction to the intermediate position for obtaining the linear behavior of airflow through the zone damper. The corrected intermediate position is determined using a mathematical model that is derived from a second fan law and based on the static pressure when the zone damper is in the fully open position, the static pressure when the zone damper is in the intermediate position, and the static pressure when the zone damper is in the fully closed position. The method includes storing the corrected intermediate position in a memory associated with the control system, wherein the corrected intermediate position that is stored in the memory is used to adjust a position of the zone damper during an operational phase of an HVAC system.

(8) These and other aspects, objects, features, and embodiments, will be apparent from the following description and the appended claims.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) The foregoing and other features and aspects of the present disclosure are best understood with reference to the following description of certain example embodiments, when read in conjunction with the accompanying drawings, wherein:

(2) FIG. 1 is an example control system of an HVAC system, in accordance with example embodiments of the present disclosure;

(3) FIG. 2 is a flowchart illustrating an example method of correcting a nonlinear behavior of airflow as a function of zone damper position of the zone dampers in the HVAC system, in accordance with example embodiments of the present disclosure;

(4) FIG. 3 is flowchart illustrating an example method of sizing the zones of the HVAC system, in accordance with example embodiments of the present disclosure;

(5) FIG. 4 is a flowchart illustrating an example method of calculating corrected zone damper positions to exhibit a linear behavior of airflow as a function of zone damper position, in accordance with example embodiments of the present disclosure; and

(6) FIG. 5 illustrates a block diagram of an example system controller of the control system of FIG. 1, in accordance with example embodiments of the present disclosure.

(7) The drawings illustrate only example embodiments of the present disclosure and are therefore not to be considered limiting of its scope, as the present disclosure may admit to other equally effective embodiments. The elements and features shown in the drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the example embodiments. Additionally, certain dimensions or positions may be exaggerated to help visually convey such principles.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

(8) The present disclosure describes a system and method to provide a linear behavior of airflow as a function of damper position of each zone damper in an HVAC system. Each zone damper of the HVAC system is incrementally closed from a fully open position to a fully closed position and static pressure measurements are recorded with each change in damper position. Then, using a mathematical model that is derived from the second fan law, a correction is calculated for each damper position of each zone damper based on the static pressure measurements to provide corrected damper positions at which the airflow through the zone damper exhibits a linear behavior. The corrected damper positions are stored. Further, during an operational cycle of the HVAC system, the corrected damper positions are applied to the zone dampers to obtain a precise airflow through the zone dampers, which in turn results in smooth temperature changes and control in a zone, as well as smooth airflow noise changes in the zone. The term 'static pressure' as used herein may generally refer to an external static pressure.

(9) Conventional HVAC systems operate under the assumption of a linear relationship of airflow

through a zone damper with each change in damper position. However, in reality the zone dampers exhibit nonlinear behavior of airflow with the change in damper positions of the zone damper. This in turn may result in control inaccuracies which affects the overall performance and control of the HVAC system. For example, in said conventional systems that assume linear airflow behavior, when there is a requirement to deliver a 20% airflow to a zone, the zone dampers of the zone may be opened by 20%. However, because of the nonlinear behavior of the airflow, the airflow through the zone may not be 20%. That is, the airflow through the zone may be more than 20%, e.g., 40%, or may be less than 20%, e.g., 5%, which in turn may cause the zone to be over-conditioned or under-conditioned. Further, the balance of airflow and pressures (e.g., static pressures) in the HVAC system may be affected and one or more zones may experience excess airflow noise and other control and performance issues. Therefore, to obtain the best performance and precise control of the HVAC system, it is desirable to correct for the nonlinearity of airflow with a change in the damper positions of a zone damper.

(10) Example embodiments of the HVAC system and method of the present disclosure will be described more fully hereinafter with reference to the accompanying drawings that describe representative embodiments of the present technology. If a component of a figure is described but not expressly shown or labeled in that figure, the label used for a corresponding component in another figure can be inferred to that component. Conversely, if a component in a figure is labeled but not described, the description for such component can be substantially the same as the description for a corresponding component in another figure. Further, a statement that a particular embodiment (e.g., as shown in a figure herein) does not have a particular feature or component does not mean, unless expressly stated, that such embodiment is not capable of having such feature or component. For example, for purposes of present or future claims herein, a feature or component that is described as not being included in an example embodiment shown in one or more particular drawings is capable of being included in one or more claims that correspond to such one or more particular drawings herein.

(11) The technology of the HVAC system and method of the present disclosure may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the technology to those appropriately skilled in the art. Further, example embodiments of the present disclosure can be located in any type of environment (e.g., warehouse, attic, garage, storage, mechanical room, basement) for any type (e.g., commercial, residential, industrial) of user.

(12) Terms such as “first”, “second”, “third”, and “within”, etc., are used merely to distinguish one component (or part of a component or state of a component) from another. Such terms are not meant to denote a preference or a particular orientation, and are not meant to limit embodiments of HVAC systems and methods of the present disclosure. In the following detailed description of the example embodiments, numerous specific details are set forth in order to provide a more thorough understanding of the invention. However, it will be apparent to one of ordinary skill in the art that the invention may be practiced without these specific details. In other instances, well-known features have not been described in detail to avoid unnecessarily complicating the description.

(13) Turning now to the figures, example embodiments of an HVAC system will be described in connection with FIGS. 1-5. In particular, a control system of an HVAC system of the present disclosure will be described in connection with FIG. 1; example operations of the HVAC system for correcting a nonlinear behavior of airflow as a function of zone damper position of the zone dampers in the HVAC system will be described in connection with FIGS. 2-4; and an example system controller of the control system will be described in connection with FIG. 5.

(14) Turning to FIG. 1, an example HVAC system **100** may include an air handler **103** that takes air from return ducts and drives the air into a plurality of supply ducts associated with distinct zones of a building. Each supply duct may include a zone damper **102** that may be controlled by a system

controller **104** to restrict or allow flow of air into each zone to achieve a desired temperature. In particular, each zone may include a zone panel **106** that may be coupled to the system controller **104** and the respective zone damper **102**. In one example embodiment, the zone panel **106** may be a simple input/output device that may be configured to adjust the damper position of the zone damper **102** based on control signals received from the system controller **104**. However, in other example embodiments, the zone panel **106** may be an intelligent device that may be configured to process information, make decisions, and perform control operations.

(15) The zone damper **102** may be a modulating damper that has one or more damper blades that may be incrementally closed. In other words, the zone damper may have several intermediate positions between a fully open position and a fully closed position. In the example embodiment of the present disclosure, the zone damper **102** may include six intermediate angular positions (herein 'intermediate positions') between the fully open position and a fully closed position. That is, the zone damper **102** may have a total of eight positions, where the first position may be an open position and the eighth position may be a closed position or vice-versa. However, one of skill in the art can understand and appreciate that in other example embodiments, the zone damper **102** may have fewer or more incremental positions between the fully open position and a fully closed position without departing from a broader scope of the present disclosure. Further, even though FIG. **1** illustrates each zone having a single zone damper **102**, one of skill in the art can understand and appreciate that in other example embodiments, each zone may have a plurality of zone dampers that are coupled together and configured to operate in concert to provide the necessary airflow to the respective zone.

(16) As illustrated in FIG. **1**, the system controller **104** may be coupled to the zone panels **106** of the different zones, the thermostats **108** associated with each zone, and the air handler **103** through a data communication bus **101** of a communication system of the HVAC system **100**, such as Rheem EcoNet™. In particular, with respect to the air handler **103**, the system controller **104** may be coupled to an air handler controller **110** that transmits air handler data to the system controller **104** and receives data from the system controller **104**. The air handler controller **110** may be configured to control a functioning of the air handler **103** in general, and/or a functioning of the different components of the air handler **103**, such as, the blower assembly **112** and the temperature control elements **114** (heating and/or cooling coils). The air handler controller **110** may be coupled to a motor **116** of the blower assembly **112** and configured to control the motor **116** based on operational requests received from the system controller **104** and/or the thermostats **108**. The motor **116** is configured to control the blades of a fan **118** to move air through the supply ducts and into the zones of the HVAC system **100** based on the operational requests. Preferably, the motor **116** may be an electronically commutated motor (ECM) and the blower assembly **112** may be a variable speed blower assembly.

(17) In one example embodiment, the system controller **104** may be any one of the thermostats **108** of the HVAC system. For example, a thermostat associated with a first zone may be configured to operate as the system controller **104**. Alternatively, in another example, a thermostat associated with the main or largest zone may be configured to operate as the system controller **104**. In other example embodiments, the system controller **104** may be a dedicated control device that is distinct from and communicatively coupled to the thermostats **108** of the different zones. In either case, the system controller **104** is configured to receive information of all the zones from the respective thermostats **108** and control the zone dampers of each zone through the respective zone panels to adjust an airflow to the respective zones. Further, the system controller **104** is configured to calculate corrections for the intermediate positions of the zone dampers **102** of the HVAC system **100** in order to achieve linear behavior of airflow with a change in the damper positions of the zone dampers **102**.

(18) One of ordinary skill in the art can understand and appreciate that in addition to the components described above, the HVAC system **100** may include many other additional

components such as filters, bypass ducts, etc. However, said additional components are not described herein to avoid obscuring the features that are associated with linearizing airflow through the zone dampers **102** of the HVAC system **100**.

(19) An example operation of the system controller **104** of the HVAC system **100** to linearize the airflow through the zone dampers for the intermediate positions of the zone dampers will be described below in greater detail in association with FIGS. 2-4.

(20) Although specific operations are disclosed in the flowcharts illustrated in FIGS. 2-4, such operations are only non-limiting examples. That is, embodiments of the present invention are well suited to performing various other operations or variations of the operations recited in the flowcharts. It is appreciated that the operations in the flowcharts illustrated in FIGS. 2-4 may be performed in an order different than presented, and that not all of the operations in the flowcharts may be performed.

(21) All, or a portion of, the embodiments described by the flowcharts illustrated in FIGS. 2-4 can be implemented using computer-readable and computer-executable instructions which reside, for example, in computer-usable media of a computer system, a memory of the system controller **104**, or like device. As described above, certain processes and operations of the present invention are realized, in one embodiment, as a series of instructions (e.g., software programs) that reside within computer readable memory of a computer system or a memory associated with the system controller **104** and are executed by the processor of the computer system or the system controller **104**. When executed, the instructions cause the computer system or the system controller **104** to implement the functionality as described below.

(22) Turning to FIG. 2, the operation **200** of the system controller **104** is executed during an initial set up phase of the HVAC system **100**. In other words, system controller **104** executes operation **200** shortly after the HVAC system **100** is installed and may or may not be repeated periodically thereafter.

(23) The operation **200** starts at step **201** and proceeds to step **202** where the system controller **104** determines the relative size of each zone of the HVAC system **100**. Determining the relative size of each zone allows the system controller **104** to further determine a share of the total system airflow that each zone may receive when the zone dampers of the HVAC system are fully open. Determining the size of each zone, as identified in step **202**, will be described below in greater detail in association with FIG. 3.

(24) Turning to FIG. 3, operation **202** associated with determining the size of each zone of the HVAC system **100** begins at step **301** where the system controller **104** turns off the temperature control elements **114** of the HVAC system **100** and opens the zone dampers **102** of all the zones. Then, in operation **302**, the system controller **104** instructs the air handler controller **110** to energize the blower assembly **112** and deliver a fixed airflow into the supply ducts and the zones of the HVAC system **100**. Responsively, in operation **303**, the system controller **104** records a static pressure (SP_{open}) across the HVAC system **100** based on the motor speed of the motor **116** that controls the fan **118** of the blower assembly **112** when the zone dampers **102** of all the zones are open.

(25) Using the motor speed, the static pressure may be obtained from a table that provides static pressure values for different motor speeds (rpm) and the resulting airflow (cfm) values. The table may be developed and stored in a memory of the system controller **104** at a factory, i.e., prior to installation of the HVAC system **100**. For example, the table is developed by subjecting the HVAC system **100** to extensive empirical testing at the factory for determining the static pressure across the HVAC system **100** for different motor speeds (rpm) and the resulting airflow (cfm) values. The process of obtaining the static pressure across the HVAC system **100** allows for operation without sensors, which may be beneficial. However, in other example embodiments, sensors may be used to determine the static pressure during operation **202**.

(26) Once the static pressure (SP_{open}) across the HVAC system **100** is determined when the zone

dampers **102** of all the zones are open, in steps **304-307**, the system controller **104**: (a) closes the zone dampers **102** of all the zones except the zone damper **102** of a first zone, and (b) instructs the air handler controller **110** to deliver the same fixed airflow as before. Responsively, the system controller **104** records a static pressure (SP_zone1) across the HVAC system **100** based on the motor speed of the motor **116** that controls the fan **118** of the blower assembly **112** when all the zone dampers **102** except the zone damper **102** of the first zone is opened. In a similar manner, sequentially, zone dampers **102** for each zone in the HVAC system **100** are opened while all other zone dampers **102** are closed. In each step of said sequence, the air handler controller **110** is instructed to deliver the same fixed airflow, and the resulting static pressure (SP_zone(i)) across the HVAC system **100** is recorded.

(27) Finally, when the static pressure (SP_zone(i)) across the HVAC system **100** for each zone that is open by itself is recorded, in operation **307**, the system controller **104** calculates the relative size of each zone of the HVAC system **100** based on the recorded static pressure values, i.e., SP_open and SP_zone(i) by using one or more of the fan laws (e.g., second fan law) and/or derivatives of the fan laws. In other example embodiments, any other appropriate mathematical models that relate the static pressure to a duct size may be used to calculate the relative size of each zone without departing from a broader scope of the present disclosure. One of skill in the art would understand how to configure the system controller **104** to compute the relative zone sizes based on the recorded static pressures using the fan laws or derivatives of the fan laws. Accordingly, the calculation of the relative zone sizes of the HVAC system **100** will not be described here in greater detail for the sake of brevity. Once the relative zone sizes are calculated, the system controller **104** reopens the zone dampers **102** of all the zones in step **308** and returns to step **203** of operation **200**.

(28) In some example embodiments, when the static pressure (SP_zone(i)) across the HVAC system **100** for each zone has been recorded, prior to calculating the relative zone sizes and returning to step **203** of operation **200**, the system controller **104** may close the zone dampers **102** of all the zones and record a static pressure (SP_closed) across the HVAC system **100** for the same fixed airflow from the blower assembly **112** to detect and determine a size of any leaks in the HVAC system **100**.

(29) Referring back to FIG. 2, in operation **203**, for each zone, the system controller **104** incrementally closes the zone damper **102** and records a static pressure (SP_position(i)) across the HVAC system **100** for each incrementally closed position of the zone damper **102**. Further, once the zone damper **102** of a respective zone reaches a fully closed position, the system controller **104** records a static pressure (SP_zone closed) across the HVAC system **100**. Then, the zone damper **102** of the respective zone is opened. Once the static pressure across the HVAC system **100** is recorded for each intermediate position and the closed position of the zone damper, in operation **204**, the system controller **104** determines, based on the recorded static pressure values, a correction for each intermediate position of the zone damper to provide a linear behavior of airflow with each change in damper position of the zone damper **102**. Determining the correction for each intermediate position of the zone damper **102** will be described below in greater detail in association with FIG. 4.

(30) Turning to FIG. 4, in operation **401**, for each intermediate position (hereinafter (position_n) of the zone damper, the system controller **104** calculates a system constant (Kn) based on the recorded values of: (a) the static pressure (SP_position(i)) across the HVAC system **100** when the zone damper is at the respective position_n, (b) the static pressure (SP_zone closed) across the HVAC system **100** when the zone damper is in the closed position, and (c) the static pressure (SP_open) across the HVAC system **100** when the zone damper is the open position. In particular, the system constant (Kn) for the position_n of the zone damper is calculated using a mathematical model comprising the following mathematical equation that is derived from the second fan law:

$$(31) \text{Kn} = \frac{(1 - \sqrt{\frac{\text{SP}_{\text{position}(n)}}{\text{SP}_{\text{zoneclosed}}}})}{(1 - \sqrt{\frac{\text{SP}_{\text{open}}}{\text{SP}_{\text{zoneclosed}}}})}$$

(32) In other words, in operation **401**, the system controller **104** applies the recorded values of the static pressures, i.e., SP_position(n), SP_open, and SP_zone closed to the above included mathematical model to generate the system constant (Kn) value for the position_n of the zone damper **102**. As described above, SP_open refers to the static pressure across the HVAC system **100** when all the zones are fully open, SP_position(i) refers to the static pressure across the HVAC system **100** for each incrementally closed position 'i' of a zone damper **102** when the other zone dampers **102** are fully open, and SP_zone closed for a zone refers to the static pressure across the HVAC system **100** when said zone is fully closed while the other zones **102** are fully open.

(33) Responsive to generating the system constant (Kn), in operation **402**, the system controller **104** calculates a correction for the system constant (Kn) associated with position_n of the zone damper. In an ideal system with a linear behavior, the value of the system constant (Kn_ideal) for each intermediate position should be equal to a value of the current intermediate position divided by the total number of damper positions of the zone damper. That is, in an ideal system with a linear behavior,

$$Kn_ideal = (\text{Damper position_n}) / (\text{Total number of damper positions})$$

(34) However, typically, the system constant (Kn) exhibits a nonlinear behavior. Therefore, the system controller **104** calculates a correction for the system constant (Kn) associated with the position_n of the zone damper **102** based on a value of the system constant (Kn_ideal) associated with the position_n in the ideal system, the value of the system constant (Kn) associated with the position_n which is calculated based on the recorded static pressure values, and the value of the system constant (K.sub.n+1) associated with the next position of the zone damper following the intermediate position_n which is calculated based on the recorded static pressure values. In particular, the correction for the system constant (Kn) associated with the position_n of the zone damper may be expressed as a percentage value and is calculated using the following mathematical equation:

$$\text{Correctionpercent}_n = \frac{(Kn_ideal - K_{n+1})}{(Kn - K_{n+1})}$$

(36) The correction for the system constant associated with the position_n of the zone damper **102** may adjust for a deviation of the system constant (Kn) from the ideal system constant (Kn_ideal) resulting from the nonlinear behavior. Responsive to calculating the correction for the system constant associated with position_n of the zone damper **102**, in operation **403**, the system controller **104** calculates a corrected position_n by applying the calculated correction for the system constant (Kn) associated with the position_n of the zone damper. In particular, the corrected position_n corresponding to the position_n of the zone damper is calculated using the following mathematical equation.

$$\text{Corrected damper position}_n = (\text{Damper position.sub.n} - (1 - \text{Correction percent.sub.n}) * (\text{Damper position.sub.n} - \text{Damper position.sub.n} + 1))$$

(37) At the corrected damper position, the system may exhibit a linear airflow behavior through the zone damper. Responsive to calculating the corrected position_n corresponding to the position_n of the zone damper **102**, in operation **404**, the system controller **104** records the corrected position_n of the zone damper **102**. Further, in operation **405**, the system controller **104** determines whether corrected positions for all the damper positions of the zone damper **102** have been calculated and recorded. If the corrected positions for all the damper positions of the zone damper **102** has not been calculated and/or recorded, then, steps **401-404** may be repeated for the remaining damper positions of the zone damper **102** till corresponding corrected positions for all the damper positions of the zone damper **102** have been calculated and recorded. Once the corresponding corrected positions for all the damper positions of the zone damper **102** have been calculated and recorded, the system controller **104** returns to step **205** of operation **200**.

(38) Returning to FIG. 2, in operation **205**, the system controller **104** checks whether the corrected positions for the intermediate positions of all the zone dampers **102** of the HVAC system **100** have been calculated. If the system controller **104** determines that corrected positions for the

intermediate positions all the zone dampers **102** have not been determined, then, step **203-204** may be repeated for the remaining zone dampers **102** of the HVAC system **100** till corrected positions for the intermediate positions of all the zone dampers **102** have been determined.

(39) After the corrected positions for the damper positions of all the zone dampers **102** have been calculated and recorded in the initial set up phase, in operation **206**, the system controller **104** may adjust the damper position of a zone damper **102** to the corrected position to deliver a specific airflow to a zone in which the zone damper **102** is disposed responsive to a demand for delivering the specific airflow to the zone. The operation **200** of the system controller **104** ends at step **207**.

(40) Operation **206** may be executed during an operational phase of the HVAC system **100** (heating or cooling cycle) to meet a demand for conditioning a zone. For example, during an operational phase of a two-zone HVAC system where 75% of the total airflow may be delivered to the first zone and 25% of the total airflow may be delivered to the second zone, the system controller **104** may determine that the airflow in the first zone has to be reduced to 20% of the normal 75% of the total airflow that is delivered to the first zone. Accordingly, in said example, the system controller **104** may adjust a zone damper **102** associated with the first zone to a corrected intermediate position of the zone damper **102** that delivers 20% of the normal 75% of the total airflow to the first zone. In conventional HVAC systems, responsive to determining that airflow in the first zone has to be reduced to 20%, the system controller **104** adjusts the zone damper **102** of the first zone to be 20% open. However, when the zone damper is 20% open, the airflow to the first zone may be more than or less than the required 20% airflow because of the nonlinear behavior of the airflow with respect to the zone damper positions. In the HVAC system **100** of the present disclosure, to achieve the 20% airflow to the first zone, the zone damper may be adjusted to the corrected position that delivers 20% airflow to the zone. The corrected position may be open more than or less than 20% based on the correction that is calculated for the nonlinear behavior of airflow through the zone damper. For example, 20% airflow may be delivered by opening the zone damper by 30% or 5%. In some example embodiments, the 20% airflow may be delivered by opening the zone damper by 20% if the airflow through the zone damper at the 20% open position exhibits a linear behavior. A linear behavior of the airflow with each change in damper position of the zone dampers allows for a precise knowledge and control of airflow to the zones, which in turn enhances the overall system performance of the HVAC system **100**.

(41) Zone sizing may be used to determine how much airflow goes through each zone when the dampers are fully open, and the linearization of the airflow may then be used to precisely adjust percentage of airflow in a specific zone. Even though FIG. 2 illustrates the zone sizing operation, i.e., operation **202** as being performed in conjunction with the damper position correction operation, i.e., operations **203-204**, one of ordinary skill in the art can understand that in some example embodiments, operation **202** may be omitted. In said example embodiments where the operation **202** of sizing the zones of the HVAC system is omitted, the static pressure associated with the fully open position of the zone dampers may be recorded as a part of operation **203**.

(42) Turning to FIG. 5, this figure illustrates an example hardware diagram of an example controller **500**. The system controller **104** may be implemented using combinations of one or more of the elements of the example controller **500**. The controller **500** includes a processor **510**, a Random Access Memory (RAM) **520**, a Read Only Memory (ROM) **530**, a memory (i.e., storage) device **540**, a network interface **550**, and an Input Output (I/O) interface **560**. The elements of the computer **500** are communicatively coupled via a bus **502**.

(43) The processor **510** comprises any well-known general purpose hardware processor. Both the RAM **520** and the ROM **530** comprise well known random access and read only memory devices, respectively, that store computer-readable instructions to be executed by the processor **510**. The memory device **540** stores computer-readable instructions thereon that, when executed by the processor **510**, direct the processor **510** to execute various aspects of the present invention described herein. As a non-limiting example group, the memory device **540** may comprise one or

more of an optical disc, a magnetic disc, a semiconductor memory (i.e., a flash based memory), a magnetic tape memory, a removable memory, combinations thereof, or any other well-known memory means for storing computer-readable instructions. The I/O interface **560** comprises input and output ports, device input and output interfaces such as a keyboard, pointing device, display, communication, and other interfaces. The bus **502** electrically and communicatively couples the processor **510**, the RAM **520**, the ROM **530**, the memory device **540**, the network interface **550**, and the I/O interface **560**, so that data and instructions may be communicated among the processor **510**, the RAM **520**, the ROM **530**, the memory device **540**, the network interface **550**, and the I/O interface **560**. In operation, the processor **510** is configured to retrieve computer-readable instructions stored on the memory device **540**, the ROM **530**, or another storage means, and copy the computer-readable instructions to the RAM **520** for execution. The processor **510** is further configured to execute the computer-readable instructions to implement various aspects and features of the present invention described herein.

(44) Although embodiments described herein are made with reference to example embodiments, it should be appreciated by those skilled in the art that various modifications are well within the scope and spirit of this disclosure. Those skilled in the art will appreciate that the example embodiments described herein are not limited to any specifically discussed application and that the embodiments described herein are illustrative and not restrictive. From the description of the example embodiments, equivalents of the elements shown therein will suggest themselves to those skilled in the art, and ways of constructing other embodiments using the present disclosure will suggest themselves to practitioners of the art. Therefore, the scope of the example embodiments is not limited herein.

Claims

1. A system in communication with a fan and a first damper that is configured to move from a first position to a second position, and from the second position to a third position, the system comprising: a controller configured to: determine a first corrective positional adjustment for the first damper for the second position at a first time interval; determine, using the first corrective positional adjustment, a fourth position for the first damper, wherein the first damper moves to the fourth position instead of the second position during a second time interval; and determine, after the first damper is in the fourth position, that airflow through the system is linear; wherein the first damper is for a first zone of the system.
2. The system of claim 1, wherein a second damper for a second zone of the system is configured to move from the first position to the second position, and from the second position to the third position, wherein the controller is further configured to: determine a second corrective positional adjustment for the second damper for the second position at a third time interval; and determine, using the second corrective positional adjustment, a fifth position for the second damper, wherein the second damper moves to the fifth position instead of the second position during a fourth time interval.
3. The system of claim 1, wherein the controller is configured to determine the first corrective positional adjustment for the first damper by: determining a first static pressure value when the first damper is in the first position; incrementing the first damper to the second position; determining a second static pressure value; and determining the first corrective positional adjustment for the first position based at least in part on the first static pressure value and the second static pressure value.
4. The system of claim 3, wherein the controller is further configured to: determine a second corrective positional adjustment for the second position based at least in part on the first static pressure value and the second static pressure value; determine a fifth position for the first damper using the second corrective positional adjustment; and cause the first damper to move to the fifth position instead of the fourth position.

5. The system of claim 3, wherein the controller is further configured to: determine a system constant value based at least in part on the first static pressure value and the second static pressure value; and determine a system constant correction value using the fourth position.
6. The system of claim 1, wherein the controller is further configured to: increment the first damper in a first direction by a first amount; increment the first damper in the first direction by a second amount; and determine that airflow through the system is linear; wherein the first corrective positional adjustment is a sum of the first amount and the second amount.
7. The system of claim 1, wherein the controller is further configured to: replace the second position with the fourth position in memory coupled to the controller.
8. The system of claim 1, wherein the first corrective positional adjustment is determined as a percent value adjustment relative to the second position.
9. The system of claim 1, wherein the controller is further configured to: determine an amount of vibration present during movement of the first damper from the first position to the second position; and determine, based at least in part on the amount of vibration, that airflow through the system is nonlinear.
10. The system of claim 1, wherein the first damper is configured to direct airflow through a heating, ventilation, and air-conditioning system.
11. A system in communication with a fan and a first damper that is configured to move from a first position to a second position, and from the second position to a third position, the system comprising: a controller configured to: determine a first corrective positional adjustment for the first damper for the second position at a first time interval; determine, using the first corrective positional adjustment, a fourth position for the first damper, wherein the first damper moves to the fourth position instead of the second position during a second time interval; and replace the second position with the fourth position in memory coupled to the controller; wherein the first damper is for a first zone of the system.
12. The system of claim 11, wherein a second damper for a second zone of the system is configured to move from the first position to the second position, and from the second position to the third position, wherein the controller is further configured to: determine a second corrective positional adjustment for the second damper for the second position at a third time interval; and determine, using the second corrective positional adjustment, a fifth position for the second damper, wherein the second damper moves to the fifth position instead of the second position during a fourth time interval.
13. The system of claim 11, wherein the controller is configured to determine the first corrective positional adjustment for the first damper by: determining a first static pressure value when the first damper is in the first position; incrementing the first damper to the second position; determining a second static pressure value; and determining the first corrective positional adjustment for the first position based at least in part on the first static pressure value and the second static pressure value.
14. The system of claim 11, wherein the first corrective positional adjustment is determined as a percent value adjustment relative to the second position.
15. The system of claim 11, wherein the controller is further configured to: determine, after the first damper is in the fourth position, that airflow through the system is linear.
16. A system in communication with a fan and a first damper that is configured to move from a first position to a second position, and from the second position to a third position, the system comprising: a controller configured to: increment the first damper in a first direction by a first amount; increment the first damper in the first direction by a second amount; and determine that airflow through the system is linear; determine a first corrective positional adjustment for the first damper for the second position at a first time interval, wherein the first corrective positional adjustment is a sum of the first amount and the second amount; and determine, using the first corrective positional adjustment, a fourth position for the first damper, wherein the first damper moves to the fourth position instead of the second position during a second time interval; wherein

the first damper is for a first zone of the system.

17. The system of claim 16, wherein a second damper for a second zone of the system is configured to move from the first position to the second position, and from the second position to the third position, wherein the controller is further configured to: determine a second corrective positional adjustment for the second damper for the second position at a third time interval; and determine, using the second corrective positional adjustment, a fifth position for the second damper, wherein the second damper moves to the fifth position instead of the second position during a fourth time interval.

18. The system of claim 16, wherein the controller is configured to determine the first corrective positional adjustment for the first damper by: determining a first static pressure value when the first damper is in the first position; incrementing the first damper to the second position; determining a second static pressure value; and determining the first corrective positional adjustment for the first position based at least in part on the first static pressure value and the second static pressure value.

19. The system of claim 16, wherein the controller is further configured to: replace the second position with the fourth position in memory coupled to the controller.

20. The system of claim 16, wherein the first corrective positional adjustment is determined as a percent value adjustment relative to the second position.
